higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T730**(E)**(A9)T
APRIL EXAMINATION

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

(8080175)

9 April 2015 (Y-Paper)
13:00–16:00

Calculators may be used.

This question paper consists of 5 pages and a formula sheet of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL the calculations must be shown.
5. ALL the sketches and diagrams must be large, clear and neat.
6. Keep questions and subsections of questions together.
7. Write neatly and legibly.

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QUESTION 1: ALTERNATING-CURRENT THEORY

1.1 The selectivity of a resonant circuit is a measure of how well a resonant circuit responds to a range of frequencies and separates others.

Indicate the difference in high and low selectivity and bandwidth by means of an amplitude-response curve versus a frequency-response curve. (5)

1.2 A circuit consists of the following components:

$Z_1 = 120 - j\ 21{,}21\ \Omega$
$Z_2 = 180 + j\ 25{,}31\ \Omega$
$Z_3 = 100 + j\ 47{,}31\ \Omega$

If Z_1 and Z_2 are connected in parallel and this combination is connected in series with Z_3 to a 250 V, 50 Hz supply, calculate:

1.2.1 The total impedance of the circuit (7)

1.2.2 The total current flow through the circuit (2)

[14]

QUESTION 2: POWER SUPPLIES

2.1 Calculate the value of the load resistor R_L if the following values of a capacitor filter are given:

V_m = 15 V; f = 50 Hz after half-wave rectification; C = 470 μf and I_{dc} = 10 mA (6)

2.2 Draw the basic circuit diagram that provides overcurrent protection in a power supply and briefly describe how the circuit operates. (6)

2.3 Calculate the drop in output voltage if a 15 V power supply (no-load voltage) has a voltage regulation of 5% as soon as it is connected to a load. (3)

[15]

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QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 Name THREE factors which cause a variation in the collector current of a transistor because of an unstable temperature. (3)

3.2 The following values of a common-emitter amplifier with a silicon transistor is known:

R_{B1} = 15,947 kΩ; R_{B2} = 3 kΩ; R_E = 120 Ω; R_c = 480 Ω; $^{+}V_{cc}$ = 12 V; V_{BE} = 0,7 V and β = 250

Calculate the values of V_e, I_c, I_B, V_{ce} and V_B of the amplifiers. (10)

3.3 Calculate the input impedance Z_i and the output impedance Zo of the circuit in QUESTION 3.2 by means of the approximate method if:

h_{ie} = 1,2 kΩ; h_{re} = 2 X 10^{-4}; h_{fe} = 100 and h_{oe} = 20 μA/V (Rs = 0) (5)
[18]

QUESTION 4: OPERATIONAL AMPLIFIERS

4.1 Explain the following terms as applied to operational amplifiers:

4.1.1 Input offset voltage

4.1.2 Inverting input

(2 × 2) (4)

4.2 4.2.1 Draw the circuit diagram of an inverting operational amplifier.

4.2.2 The input signal to an operational amplifier in the inverting mode is -2 V and the input resistance is 20 kΩ.

If the output voltage is 20 V, calculate the feedback resistor.

(2 × 3) (6)

4.3 Draw a neat, labelled circuit diagram of an operational band-reject filter. (4)
[14]

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QUESTION 5: INTEGRATED CIRCUITS

5.1 What should be done with the unused inputs of a CMOS-integrated circuit when it is connected to a circuit? (2)

5.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (5.2.1–5.2.3) in the ANSWER BOOK.

5.2.1 CMOS-integrated circuits have higher noise immunity.

5.2.2 CMOS-integrated circuits are susceptible to static charges because of their low reactive input.

5.2.3 As soon as work is done on the circuit of CMOS-integrated circuits, the power supply to the circuit must be switched off.

(3 × 1) (3)
[5]

QUESTION 6: TRANSDUCERS

6.1 Draw a neat, labelled circuit diagram of a Schmitt-trigger which includes a photo-transistor that switches a relay on and off. (6)

6.2 Briefly explain the operating principle of a closed-loop system that makes use of two potentiometers in order to control a canon. (5)
[11]

QUESTION 7: ELECTRONIC PHASE CONTROL

Draw a neat, labelled circuit diagram of a light dimmer which makes use of a triac. **[6]**

QUESTION 8: TEST EQUIPMENT

Draw a neat, labelled block diagram of a dual slope A/D-voltmeter. **[6]**

QUESTION 9: OSCILLATORS

9.1 State THREE factors that may destabilise the frequency of an oscillator. (3)

9.2 Draw a neat, labelled circuit diagram of a uni-junction transistor oscillator. (4)

9.3 Name TWO common types of waveforms and give an application of each. (4)
[11]

TOTAL: 100

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INDUSTRIAL ELECTRONICS N5

FORMULA SHEET

$$I = \frac{V}{R} \qquad P = IV = I^2R = \frac{V^2}{R}$$

$$V_T = V_1 + V_2 + V_3 + ... = I_1R_1 + I_2R_2 + I_3R_3 + ...$$

$$I_T = I_1 + I_2 + I_3 + ... = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + ...$$

$$T = RC \qquad T = \frac{L}{R}$$

$$V_R = RC\frac{dv}{dt} \qquad V_C = \frac{1}{RC}\int v_i dt$$

$$X_L = 2\pi fL \qquad X_C = \frac{1}{2\pi fC}$$

$$Z = R + jX_L \qquad Z = R - jX_C$$

$$Z = R + j(X_L - X_C) \qquad I_T = \frac{V_T}{Z_T}$$

$$V_R = I_TR \qquad V_L = I_T(jX_L)$$

$$V_C = I_T(-jX_C) \qquad f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{V_L}{V_T} = \frac{V_C}{V_T} = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{f_2 - f_1}$$

$$BW = f_2 - f_1 \qquad \frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$Z_T = \frac{Z_2Z_2}{Z_1 + Z_2} \qquad I_T = I_1 + I_2 = \frac{V}{Z_1} + \frac{V}{Z_2}$$

$$Z_T = \frac{R(jX_L)}{R + jX_L} \qquad \frac{1}{Z_T} = \frac{1}{R} - \frac{j}{X_L}$$

$$I_T = I_R - jI_L \qquad I_T = \frac{V}{R} - j\frac{V}{X_L}$$

$$Z_T = \frac{R(-jX_C)}{R - jX_C} \qquad \frac{1}{Z_T} = \frac{1}{R} + \frac{j}{X_C}$$

$$I_T = I_R + jI_C \qquad I_T = \frac{V}{R} + j\frac{V}{X_C}$$

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$$\frac{1}{Z_T} = \frac{1}{R} - j\left(\frac{1}{X_L} - \frac{1}{X_C}\right)$$

$$I_T = I_R - j(I_L - I_C)$$

$$I_T = \frac{V}{R} - j\left(\frac{V}{X_L} - \frac{V}{X_C}\right)$$

$$a + jb = \sqrt{a^2 + b^2} \,/\, \tan^{-1}\frac{b}{a} = r \,\underline{/\theta}$$

$$r\,\underline{/\theta} = r(\cos\theta + j\sin\theta)$$

$$Q = \tan\theta$$

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$Z_d = \frac{L}{CR_1}$$

$$V_{rms} = \frac{1}{\sqrt{2}}V_m = 0{,}707\,V_m$$

$$V_{dc} = \frac{2}{\pi}V_m = 0{,}637\,V_m$$

$$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P}$$

$$V_{dc} = \frac{1}{\pi}V_m = 0{,}318\,V_m$$

$$PIV = V_m$$

$$PIV = 2\,V_m$$

$$R_{r(rms)} = 0{,}385\,V_m$$

$$V_{r(rms)} = 0{,}305\,V_m$$

$$r = \frac{V_{r(rms)}}{V_{dc}}$$

$$V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

$$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$$

$$V_{dc} = V_m - \frac{I_{dc}}{2\,fC}$$

$$V_{dc} = V_m - \frac{I_{dc}}{4\,fC}$$

$$V_{r(rms)} = \frac{I_{dc}}{2\sqrt{3}\,fC} = \frac{V_{dc}}{2\sqrt{3}\,fCR_L}$$

$$V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3}\,fC} = \frac{V_{dc}}{4\sqrt{3}\,fCR_L}$$

$$r = \frac{I_{dc}}{2\sqrt{3}\,fCV_{dc}} = \frac{1}{2\sqrt{3}\,fCR_L}$$

$$r = \frac{I_{dc}}{4\sqrt{3}\,fCV_{dc}} = \frac{1}{4\sqrt{3}\,fCR_L}$$

$$V'_{dc} = \frac{R_L}{R + R_L}.V_{dc}$$

$$V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}}.V_{r(rms)}$$

$$X_C = \frac{1}{2\pi fC} \qquad X_C = \frac{1}{4\pi fC} \qquad r' = \frac{V'_{r(rms)}}{V'_{dc}}$$

$$V'_{r(rms)} = \frac{X_C}{R}.V_{r(rms)}$$

$$r' = rX_C\left(\frac{R + R_L}{R.R_L}\right)$$

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$$V'_{dc} = V_{dc} - I_{dc}R_1$$

$$V'_{dc} = \frac{R_L}{R_L + R_1} . V_{dc}$$

$$V'_{r(rms)} = \frac{V_{r(rms)}}{(2\pi f)^2 LC}$$

$$V'_{r(rms)} = \frac{V_{r(rm)}}{(4\pi f)^2 LC}$$

$$VR = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

$$2V_m = V_{c2} = V_m + V_{c1}$$

$$3V_m = V_{c1} + V_{c3} = V_m + 2V_m$$

$$S = \frac{\Delta V_o}{\Delta V_i}$$

$$V_R = V_i - V_z$$

$$R_{s(\min)} = \frac{V_{i(\max)} - V_z}{I_{z(\max)}}$$

$$I_z = \frac{P_z}{V_z}$$

$$R_{L(\min)} = \frac{V_Z}{V_{i(\max)} - V_Z} . R_S$$

$$V_o = V_r - V_{be}$$

$$R_c = \frac{V_{cc} - V_{ce}}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be}}{I_b}$$

$$\beta = \frac{I_c}{I_b}$$

$$C_e \geq \frac{10}{2\pi f R_e}$$

$$V_e = \frac{V_{cc}}{10}$$

$$R_e = \frac{V_e}{I_e} \simeq \frac{V_e}{I_c}$$

$$R_c = \frac{V_{cc} - V_{ce} - V_e}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be} - V_e}{I_b}$$

$$R_{b1} = \frac{R_{b2}(V_{cc} - V_b)}{V_b}$$

$$R_{b2} = \frac{1}{10}\beta R_e$$

$$V_b = V_e + V_{be}$$

$$V_{be} = h_{ie} i_b + h_{re} V_{ce}$$

$$i_c = i_{fe} i_b + h_{oe} V_{ce}$$

$$A_i = \frac{h_{fe}}{1 + h_{oe} Z_L}$$

$$A_i = h_{fe}$$

$$A_i = \left(\frac{h_{fe}}{1 + h_{oe} Z_L}\right)\left(\frac{R_b T}{R_{bT} + Z_1}\right)\left(\frac{R_c}{R_c + R_L}\right)$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re}) Z_L}$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie}}$$

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$$Z_1 = h_{ie} - \frac{h_{fe} h_{re} Z_L}{1 + h_{oe} Z_L} \qquad Z_1 = h_{ie}$$

$$Z_2 = \frac{1}{h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_s}} \qquad Z_2 = \frac{1}{h_{oe}}$$

$$A_p = \frac{A_i^2 R_L}{R_1} = -A_v A_i \qquad A_p = \frac{h_{fe}{}^2 R_L}{h_{ie}}$$

$$Z_0 = R_C // R_L // Z_2 = Z_L // Z_2$$

$$Z_0 = R_C // Z_2 = Z_L // Z_2 \qquad Z_1 = R_b // Z_1$$

$$Z_i = R_{b1} // R_{b2} // Z_1 \qquad I_1 = \frac{R_{bT} I_i}{R_{bT} = Z_1}$$

$$I_0 = h_{fe} I_b = h_{fe} \left(\frac{R_{b2}(I_i)}{R_{b2} + h_{ie}} \right) \qquad A_i = \frac{I_0}{I_1}$$

For common base, substitute all the 'e' subscripts with a 'b' in the h-parameters.

$$Z_L = R_c // R_L \qquad I_1 = \frac{R_e I_i}{R_e + Z_1}$$

$$CMRR = \frac{A_{dm}}{A_{cm}} \qquad CMRR\ (dB) = 20log \frac{A_{dm}}{A_{cm}}$$

$$I_e = \frac{V_e}{R_e} \qquad I_c = \frac{I_e}{2}$$

$$R_L = \frac{V_{R_L}}{I_C} \qquad g_m R_L = \frac{h_{fe}}{h_{ie}} . R_L$$

$$V_0 = -\left(\frac{R_f}{R_1} \right) . V_i \qquad V_0 = \left(\frac{R_f}{R_1} + 1 \right) . V_i$$

$$V_0 = -\left(\frac{R_f}{R_1} . V_1 + \frac{R_f}{R_2} . V_2 + \frac{R_f}{R_3} . V_3 \right) \qquad V_0 = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) R_f$$

$$V_0 = -(V_1 + V_2 + V_3) \qquad V_0 = -(I_1 + I_2 + I_3)\, R_f$$

$$V_0(t) = -\frac{1}{RC} \int V_i(t) \qquad V_0(t_b) = -\frac{1}{RC} \int_{t_a}^{t_b} V_i(t_b) + V_c(t_a)$$

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$t = \frac{1}{f}$

$A_v = -\frac{R_s}{R_1}$

$R_2 = \frac{R_1 R_s}{R_1 + R_s}$

$f_c = \frac{1}{2\pi R_s C}$

$V_0(t) = -RC\frac{dV_i(t)}{dt}$

$A = \frac{R_f}{X_c}$

$A = -\frac{R_f}{R_s}$

$V_0(t) = -R_f C\frac{d}{dt}.v_i \sin\omega t$

$t = R_f C$

$V_0 = A(V_r - V_i)$

$V_0 = \frac{R_f}{R_s}(V_2 - V_1)$

$V_0 = V_2 - V_1$

$f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}}$

$f_0 = \frac{1}{2\pi RC}$

$f_0 = \frac{1}{2\pi\sqrt{L_T C_1}}$

$L_T = L_1 + L_2 + 2M$

$f_0 = \frac{1}{2\pi\sqrt{LC_T}}$

$C_T = \frac{C_1 C_2}{C_1 + C_2}$

$f_0 = \frac{1}{2\pi\sqrt{LC_2}}$

$f = \frac{1}{2\pi RC\sqrt{6}}$

$f_0 = \frac{1{,}5}{RC}$

$f_0 = \frac{1}{t} = \frac{1}{t_1 + t_2}$

$t_1 = 0{,}7\ R_2 C_1$

$t_2 = 0{,}7\ R_1 C_2$

$f_0 = \frac{1}{1{,}4RC}$

$V_i = I_{c2}R_e + V_{be(ON)}$

$t = 1{,}1\ RC$

$f_0 = \frac{1{,}443}{(R_A + 2R_B)C}$

$t_{low} = 0{,}693(R_B)C$

$t_{high} = 0{,}693(R_A + R_B)C$

$t_T = t_{low} + t_{high}$

$K = \frac{\Delta R / R}{\Delta l / l}$

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$$\sigma = \Delta l / l \qquad R = \rho \frac{1}{\pi d^2 / 4}$$

$$\sigma = \frac{S}{E} \qquad Resolution = \frac{1}{amount\ of\ turns}$$

$$Resolution = \frac{voltage\ drop\ across\ adjacent\ turns}{total\ voltage\ drop}$$

$$R_t = Ae^{B/T} \qquad T = 273 + °C$$

$$V_A = \frac{R_2}{R_1 + R_2}.V_T \qquad V_B = \frac{R_t}{R_t + R_3}.V_T$$

$$V_{AB} = V_A - V_B \qquad A_v = \frac{V_0}{V_i}$$

$$V_{Hall} = kIH$$

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